Computer Science & IT

Database Management System

Relational Model & Normal Forms

Lecture No. 06

By- Vishal Sir

Recap of Previous Lecture

Topic Super key

{Calculate # super keys based on given C.Ks}

Topic Closure of an attribute set

$(X)^+$ = Set of all attributes which can be determined by X

Topic Candidate key identification w.r.t. given set of FDs

Slide

Topics to be Covered

Topic Candidate key identification

Slide

Topic : Candidate key (Minimal Super key)

- Let R be the relational schema, and let X be the super key of relation R.
  If no proper subset of X is a super key then X is minimal super key i.e., X is Candidate key

eg. let $(AB)^+$ = All attributes of relation R, ∴ (AB) is a Superkey of R

Non-Empty proper Subsets of AB are → {A}, if $(A)^+$ = Not all attributes of R

and

→ {B}, if $(B)^+$ = Not all attributes of R

⇒ i.e. No Proper subset of (AB) is a S.K.

Hence, (AB) will be a Candidate Key

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

AB→C,

B→E,

C→D

Here A & B are essential attributes of the relation

Find the Candidate key of R.

The attributes of the relation which are not present in the R.H.S. Part of any FD are called "Essential attributes" of the relation

Note: Every essential attribute must be present in every Candidate key of the relation

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

AB→C,

B→E,

C→D

Find the Candidate key of R.

$(AB)^+ = \{\underbrace{A, B, C, D, E}_{\text{all attributes}}\}$ ∴ AB is a S.K

* Check for C.K

Proper Subsets of (AB)

$(A)^+ = \{A\}$ Not all attributes ∴ "A" is not a S.K.

$(B)^+ = \{B, E\}$ Not all attributes, ∴ B is not a S.K

None of the proper subset of (AB) is a S.K.
∴ $\boxed{AB}$ is a Candidate key

$(AB)^+ = \{\underbrace{A, B, C, E, D}_{\text{all attributes of Relation}}\}$

∴ AB is a Super key of relation

{ we know Both A & B are Essential
∴ No need to check w.r.t. Proper subsets of AB.

and

AB is a Candidate key.

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

$AB \rightarrow C$,

$B \rightarrow E$,

$C \rightarrow D$

Find the Candidate key of R.

$\boxed{AB}$ is a Candidate Key of the relation

$\therefore$ Prime attributes = $\{A, B\}$

observe that, None of the prime attribute is present in R.H.S. part of any FD of the given FD set

Hence $\boxed{AB}$ will be the only Candidate Key of given Relⁿ.

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

AB→C,

B→E,

C→D

E→A

"B" is not present in RHS. part of any FD

∴ "B" is an Essential attribute.

$$(B)^+ = \underbrace{\{B, E, A, C, D\}}_{\text{all attributes}}$$

∴ B is a Super Key.

"B" is a key with single attribute

∴ $\boxed{B}$ is a Candidate key

Find the Candidate key of R.

Prime Attribut = $\underbrace{\{B\}}$

Not present in RHS of any FD

∴ $\boxed{B}$ is the only CK.

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

$AB \rightarrow C$,

$B \rightarrow E$,

$C \rightarrow D$

$E \rightarrow A$

Find the Candidate key of R.

→ $(AB)^+ = \underbrace{\{A, B, C, D, E\}}_{\text{all attributes}}$ ∴ AB is a Super Key

• Check for Candidate Key

Proper subsets of AB

$(A)^+ = \{A\}$ Not all attributes ∴ Not a S.K

$(B)^+ = \underbrace{\{B, E, A, C, D\}}_{\text{All attributes}}$ ∴ $\underbrace{\text{B is a S.K}}_{\text{One of the Proper Subset of AB is a S.K.}}$

∴ AB is just a S.K, it is not a CK

B is a Super Key

→ Now Check if B is a C.K. or not

"B" is a key with a single attribute

∴ $\boxed{B}$ is a Candidate Key

Prime Attribute = {B}

Topic : Note

If there excists any non-trivial FD in the relation of the form $X \rightarrow Y$

such that Y is a prime attribute in the relation.

then, relation will have more than one Candidate Key

{ To find those Candidate Keys we will replace Y in the Candidate Key by X and we will get a new super key, and w.r.t that new super key we will try to find new Candidate key }

#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

AB→CD,

D→A

Find all the Candidate keys of R.

$(AB)^+ = \{\underbrace{A, B, C, D}_{\text{all attributes of relation}}\}$, ∴ AB is a S.K.

Check for Candidate key.

Proper subsets of AB

$(A)^+ = \{A\}$ Not all attributes

$(B)^+ = \{B\}$ Not all attributes

No proper subset of (AB) is a Super key

∴ $\boxed{AB}$ is a C.K.

B is the Essential attribute

$(B)^+ = \{\underbrace{B}_{\text{Not all attributes}}\}$

∴ B is not a S.K,

but B will be present in Every Candidate Key

$\boxed{AB}$ is a Candidate Key

Prime Attributes = $\{A, B\}$

We have FD $D \rightarrow A$

s.t. 'A' is prime attribute

$\therefore$ Replace "A" by L.H.S. part of FD

i.e., Replace A by D

Replace A by D

DB, is a super key

Check for C.K

$(D)^+ = \{D, A\}$

$(B)^+ = \{B\}$

No proper subset of DB is a S.K

$\boxed{DB}$ is a Candidate Key

Prime Attributes = $\{A, B, D\}$

$AB \rightarrow CD$

$\equiv AB \rightarrow C$ & $AB \rightarrow D$

$AB \rightarrow$ "D"

$\therefore$ Replace D by AB

No other replacement is possible,

$\boxed{AB}$ and $\boxed{DB}$ are the only Candidate Keys of the relation.

#Q. Assume a relation R (M, N, O, P, Q) that has the following functional dependencies:

MNO →PQ and

P → MN

Find the Candidate keys of R.

"O" is the essential attribute

$(O)^+ = \underbrace{\{O\}}_{\text{Not all attributes}}$

∴ "O" alone can not be a C.K.

$(MNO)^+ = \{\underbrace{M, N, O, P, Q}_{\text{all attributes}}\}$

∴ (MNO) is a S.K

Check for candidate key

$(MO)^+ = \{M, O\}$

$(NO)^+ = \{N, O\}$

No proper subset of (MNO) is a S.K

∴ $\boxed{MNO}$ is a Candidate key.

MNO is a candidate key

Prime attributes = {M, N, O}

P → MN
both are prime attributes

Replace "MN" by P

PO is a Super key

Check for Candidate key

$(P)^+ = \{P, M, N\}$
$(O)^+ = \{O\}$
No proper subset is a S.K.

PO is a C.K.

Prime Attributes = {M, N, O, P}

MNO → P
∴ Replace "P" by "MNO"

P → MN
∴ P → M

MNO is a Candidate key

Prime attributes = {M, N, O}

∴ Replace M by P

PNO is S.K

Check for C.K.

$(NO)^+ = \{N, O\}$

$(PO)^+ = \underbrace{\{P, O, M, N, Q\}}_{\text{all attributes}}$

∴ (PO) is S.K

Hence (PNO) Can not be a C.K

Check whether (PO) is a C.K or not

P → MN
∴ P → N
∴ Replace N by P

MPO is S.K

Check for C.K

$(MO)^+ = \{M, O\}$

$(PO)^+ = \underbrace{\{P, O, M, N, Q\}}_{\text{all attributes}}$

∴ (PO) is S.K

Hence (MPO) Can not be a C.K

Check whether (PO) is a C.K or not

We will find

$\boxed{PO}$ is a C.K

Prime Attributes = {M, N, O, P}

Replace "P" by "MNO"

#Q. Assume a relation R (A, B, C, D) that has the following functional dependencies:

$AB \to CD$

$C \to A$

$D \to B$

Find the Candidate keys of R.

No Essential attribute

$(AB)^+ = \{\underbrace{A, B, C, D}_{\text{all attributes}}\}$

$\therefore$ AB is a Super Key

Check for Candidate Key

$(A)^+ = \{A\}$

$(B)^+ = \{B\}$

No Super Key

$\therefore$ $\boxed{AB}$ is a C.K.

$\boxed{AB}$ is a Candidate Key

Prime attributes = {A, B}

$C \to A$

∴ Replace A by C

CB is S.K

Check for C.K

$(C)^+ = \{C, A\}$

$(B)^+ = \{B\}$

$\boxed{CB}$ is a C.K

Prime Attribute = {A, B, C}

$C \to A$ & $D \to B$

∴ Replace A by C & B by D

$D \to B$

∴ Replace B by D

AD is S.K.

Check for C.K

$(A)^+ = \{A\}$

$(D)^+ = \{D, B\}$

∴ $\boxed{AD}$ is a C.K.

P.A = {A, B, D}

$AB \to CD$

∴ Replace CD by AB ∴

$D \to B$

∴ Replace B by D

CD is S.K

Check for C.K

$(C)^+ = \{C, A\}$

$(D)^+ = \{D, B\}$

∴ $\boxed{CD}$ is C.K P.A = {A, B, C, D}

$C \to A$

∴ Replace A by C

(AB), (AD), (CB) & (CD) are the four Candidate keys of the relation

#Q. Assume a relation $\underbrace{R\,(A, B, C, D, E, H)}$ that has the following functional dependencies:

$$\left\{\begin{array}{l} A \rightarrow B \\ BC \rightarrow D \\ E \rightarrow C \\ D \rightarrow A \end{array}\right.$$

Find the Candidate keys of R.

- E & H are the essential attributes
- $(EH)^+ = \{E, H, C\}$

E & H will be present in Every C.K

& "C" Can be determined using E & H

∴ "C" will not be required in any C.K.

$(AEH)^+ = \{\underbrace{A, E, H, C, B, D}_{\text{all attributes}}\}$

∴ AEH is a Super key.

as well as

$\boxed{AEH}$ is a Candidate key.

$\boxed{AEH}$ is a Candidate Key

Prime Attributes = {A, E, H}

$D \to A$

$\therefore$ Replace A by D

DEH is a Super Key

as well as $\boxed{DEH}$ is a Candidate Key

Prime Attributes = {A, E, H, D}

$BC \to D$

$\therefore$ Replace D by BC

BCEH is a S.K $\Rightarrow$ Check for Candidate Key

$(BEH)^+ = \{\underbrace{B, E, H, C, D, A}_{\text{all attributes}}\}$

$\therefore$ BEH is a S.K.,

Hence "BCEH" is just a Super Key, it is not a C.K.

$(CEH)^+ = \{C, E, H\}$

BEH is a new (smaller size) Super Key

$\boxed{BEH}$ is a Candidate Key

Prime Attributes = {A, E, H, D, B}

$A \to B$

$\therefore$ Replace B by A

(AEH), (DEH) & (BEH) are the Candidate Keys of the relation

2 mins Summary

Topic Candidate key identification

Slide

THANK - YOU

Slide